

**Academic Positions:**

Assistant Professor, Faculty of Economics, University of Cambridge, 2024 –

**Past Academic Positions:**

Assistant Professor of Economics, University of Tokyo, 2022 – 2024

Visiting Scholar, University of Cambridge, March 2023

Postdoctoral Research Associate, University of Cambridge, 2021 – 2022

Dissertation Fellow, Federal Reserve Bank of San Francisco, June 2019 – August 2019

**Education:**

Ph.D., Economics, University of Pennsylvania, 2021

(Advisors: Jesús Fernández-Villaverde and Dirk Krueger,

Committee: Frank Schorfheide and Andrew Abel)

M.A., Economics, University of Pennsylvania, 2021

M.S., Finance, University of Illinois at Urbana-Champaign, 2016

B.B.A., Business Administration, Korea University, 2014

B.S., Mathematics, Korea University, 2014

**Research Fields:**

Macroeconomics, Finance

**Publications:**

*“Bridging Micro and Macro Production Functions: The Fiscal Multiplier of Infrastructure Investment”* with Minsu Chang

- *Journal of Monetary Economics*, Volume 157, January 2026

*“The Risk-Premium Channel of Uncertainty: Implications for Unemployment and Inflation”*  
with Lukas B. Freund and Pontus Rendahl

- *Review of Economic Dynamics*, Volume 51, December 2023, Pages 117-137

**Working Papers:**

*“Striking While the Iron Is Cold: Fragility after a Surge of Lumpy Investments”*

- Accepted at *Journal of the European Economic Association*

*“A Global Dynamic Nonlinear Solution Framework and the Repeated Transition Method”*

- R&R at *The Review of Economic Studies*

*“Disclosure Regulation, Intangible Capital and the Disappearance of Public Firms”*

with Sara Casella and Sergio Villalvazo

*“An Analytic Theory of Frictional Firm Dynamics”* with Vasco M. Carvalho

*“Dancing on the Saddles: A Geometric Framework for Stochastic Equilibrium Dynamics”*

*“Dry Firms, Deep Recessions: Corporate Payouts and Aggregate Dynamics”*

with Miguel H. Ferreira and Timo Haber

*“The Convex Origin of Fixed Costs”*

*“Endogenous Separations and Non-monotone Beveridge Curve Shifts”*

with Philip Schnattinger and Francesco Zanetti

*“The Spender of Last Resort: Global Equilibrium Dynamics under the Zero Lower Bound”*

with Kao Nomura

*“Endogenous Plucking Through Networks: The Plucking Paradox”* with Ye Sun

*“Timing Search”*

### **Work in Progress:**

*“Top Income Inequality and the Business Cycle”*

*“Rising Intangibles and Technology-Biased Technical Change”* with Jesús Fernández-Villaverde

*“From Spreads to Spirals: How Financial Frictions Drive Lumpy Investments”*

with Miguel H. Ferreira and Timo Haber

*“Tracing Data Through Analytical General Equilibrium”*

with Minji Bang

### **Presentations (\*scheduled):**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2027 | Durham University*                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2026 | University of London (Birkbeck), University of Manchester, Aarhus University, University of Exeter, E1 Macro Workshop at QMUL, University of Copenhagen, Summer Conference on Macro at CityUHK, BSE Summer Forum*, SED Athens*, Nanyang Technological University*, Singapore Management University*, Seoul National University*                                                                                                                                                              |
| 2025 | Chung-Ang University, Workshop on AI and New Methods in Macro (University of Zurich), Cowles Conference on Macroeconomics (Yale University), Workshop on Macroeconomics and Productivity (University of Alicante), BSE Summer Forum, RES Annual Conference, World Congress of the Econometric Society, Korea University, SITE (Stanford University), Kyoto University, Jonkoping University, Bank of England, London Junior Macro Workshop, University of Glasgow, ANU $T_v=v$ reading group |
| 2024 | University of Tokyo, KER International Conference, SED Barcelona, Yonsei Macroeconomics Workshop, KDI Summer Workshop in Macroeconomics, Dynare conference (NUS), Korea University, KAEA-ASSA/AEA                                                                                                                                                                                                                                                                                            |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023 | Osaka University, University of Sydney, Monash University, University of Melbourne, Queen Mary University of London, Tinbergen Institute Amsterdam, Singapore Management University, National University of Singapore, University of Cambridge, Canadian Economics Association Conference, HKUST/Jinan Macro Workshop, East Asia Macroeconomics Mini-Conference (The Chinese University of Hong Kong), City University of Hong Kong                                                                                                                                                                                                                                              |
| 2022 | Bank of England, Yonsei University, Canadian Economics Association Conference, KEA International Conference, Society for Computational Economics (CEF), RES Junior Symposium, Econometric Society Australasia Meeting, Asian Meeting of the Econometric Society in East and South-East Asia, EEA-ESEM, Korea Development Institute, Hitotsubashi University, ITAM-PIER Conference on Macro-Finance, The Chinese University of Hong Kong, University of Hong Kong, City University of Hong Kong, Waseda University, Keio University, University of Colorado Boulder, Midwest Macro Conference (Fall), National Graduate Institute for Policy Studies (GRIPS), University of Tokyo |
| 2021 | AEA/ASSA (Virtual), Korea Development Institute, Bilkent University, Federal Reserve Board, Bank of Korea, Federal Reserve Bank of St. Louis, Federal Reserve Bank of Boston, University of Tokyo, University of Florida, Korea University (Business), Korea University (Econ), SED Minneapolis, MMF Annual Conference, University of Cambridge, European Winter Meeting of the Econometric Society                                                                                                                                                                                                                                                                              |
| 2020 | KEA International Conference, WEAI Annual Meeting, MEA Annual Meeting (cancelled), University of Pennsylvania                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2019 | Federal Reserve Bank of San Francisco, University of Pennsylvania                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

#### **Discussions (\*scheduled):**

|      |                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | “The Macroeconomic Effects of Cash Transfers: Evidence from Brazil” by Arthur Mendes, Wataru Miyamoto, Thuy Lan Nguyen, and Steven Pennings, Workshop Oxford-CEPR-BoJ-Waseda             |
| 2023 | “Evergreening” by Miguel Faria-e-Castro, Pascal Paul, and Juan M. Sánchez, Workshop on Financial Frictions, Zombie Firms and the Macroeconomy at the University of Oxford                |
| 2022 | “Understanding Trend Inflation Through the Lens of the Goods and Services Sectors” by Yunjong Eo, Luis Uzeda, and Benjamin Wong, Japan-Korea Economic Society Academic Exchange Workshop |

#### **Professional Activities:**

|                      |                                                                                                                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conference organizer | GRIPS-UT Macroeconomics & Policy Workshop in Tokyo, 2024                                                                                                                                                           |
| Referee              | Journal of the European Economic Association, The Review of Economics and Statistics, International Economic Review, Journal of Economic Dynamics and Control, Games and Economic Behavior, Macroeconomic Dynamics |

### **Honors, Scholarships, Research Grants, and Fellowships:**

|           |                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------|
| 2025      | Keynes Fund Research Grant, University of Cambridge (~ £50,000)                                                                    |
| 2024      | Keynes Fund Research Grant, University of Cambridge (~ £10,000)                                                                    |
| 2023      | Grants-in-Aid for Scientific Research, Kakenhi, Japan Society for the Promotion of Science: Early-Career Scientists (~ ¥4,420,000) |
| 2022      | Best Paper Award for Early Career Academics at Econometric Society Australasia Meeting                                             |
| 2020      | Hiram C. Haney Fellowship Award in Economics for Best Third Year Research Paper                                                    |
| 2019      | FRB San Francisco Thomas J. Sargent Dissertation Fellowship                                                                        |
| 2018      | Princeton Initiative Summer Program                                                                                                |
| 2016-2020 | University of Pennsylvania Doctoral Fellowship                                                                                     |
| 2016-2020 | Kwanjeong Educational Foundation Fellowship                                                                                        |
| 2015      | American Finance Association (AFA) Doctoral Student Travel Grant Award                                                             |
| 2014      | Zwisler Fellowship (merit-based), University of Illinois at Urbana-Champaign                                                       |
| 2014-2015 | University of Illinois at Urbana-Champaign Doctoral Fellowship                                                                     |

### **Teaching Experience:**

|                            |                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| Michaelmas, 2024, 2025     | University of Cambridge, <i>Computational Methods for Macroeconomics</i> (Ph.D.)                                  |
| Michaelmas, 2024, 2025     | University of Cambridge, <i>Advanced Macroeconomics I</i> (MPhil in Research)                                     |
| Spring, 2023               | University of Tokyo, <i>Macroeconomics I</i> (Ph.D., core)                                                        |
| Spring, 2023, Fall, 2022   | University of Tokyo, <i>Macroeconomy and Capital Markets</i> (Ph.D.)                                              |
| Lent, 2022                 | University of Cambridge, <i>Computational Methods for Macroeconomics</i> (Ph.D.)                                  |
| Summer, 2019, Summer, 2020 | University of Pennsylvania, <i>Math Camp</i> (Ph.D.)                                                              |
| Spring, 2018               | University of Pennsylvania, <i>Intermediate Macroeconomics</i> , Teaching Assistant to Professor Dirk Krueger     |
| Fall, 2017                 | University of Pennsylvania, <i>Econometrics I: Fundamentals</i> (Ph.D.), Teaching Assistant to Professor Xu Cheng |
| Spring, 2013               | Korea University, <i>Investments</i> , Teaching Assistant to Professor Baeho Kim                                  |

**Languages:** English (fluent), Korean (native)

**Computational Skills:** MATLAB, Julia, R, Stata, SAS